Resilience Academy

SDI Training for ICT Professionals

QGIS Map Making

Styling, Labelling, and Print-Ready Maps

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Day 4 | 30 Minutes Lecture + Extended Practical | crd.resilienceacademy.ac.tz

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Styling Layers: Categorised and Graduated Symbols

Categorised Symbols

Use when your data has **distinct categories** (e.g., facility type: school, hospital, market).

1. Right-click layer → **Properties**
2. Go to **Symbology** tab
3. Change "Single Symbol" to **Categorized**
4. Choose the **Column** (e.g., `facility_type`)
5. Click **Classify** — each category gets a unique colour
6. Click **OK**

Graduated Symbols

Use when your data has **numeric values** (e.g., number of students, flood depth).

1. Right-click layer → **Properties**
2. Go to **Symbology** tab
3. Change to **Graduated**
4. Choose the **Column** (e.g., `num_students`)
5. Set number of **Classes** (e.g., 5)
6. Choose a **Color ramp** (e.g., green to red)
7. Click **Classify**, then **OK**

The Attribute Table: View, Filter, Select

Opening the Attribute Table

- Right-click on a WFS layer → **Open Attribute Table**
- You see a spreadsheet of all features and their properties
- Each **row** = one feature (one school, one point)
- Each **column** = one attribute (name, type, location)

Selecting Features

- Click a row to **select** it — it highlights on

Filtering with Expressions

Example: find all education facilities in Ilemela district.

1. Open Attribute Table
2. Click the **Select by Expression** icon (yellow square with ε)
3. Type: `"district" = 'Ilemela'`
4. Click **Select Features**
5. Selected features turn **yellow** on the map

Remember

Adding Labels to Your Map

How to Add Labels

1. Right-click layer → **Properties**
2. Go to the **Labels** tab
3. Change "No Labels" to **Single Labels**
4. Choose the **Value** column:
   - For wards: select `ward_name`
   - For facilities: select `facility_name`
5. Click **OK** — names appear on the map

Improving Label Appearance

In the Labels properties, you can adjust:

- **Font** — size, colour, bold/italic
- **Buffer** — white outline around text for readability (enable under *Buffer* sub-tab)
- **Placement** — above, below, or centred on the feature
- **Rendering** — show labels only at certain zoom levels

Best Practice

Print Layout: Creating a Professional Map

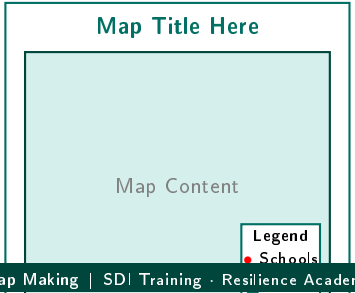
Opening the Print Layout

1. Go to **Project** → **New Print Layout**
2. Give it a name (e.g., "Mwanza Education Map")
3. A blank page opens — this is your map canvas for printing

Adding Map Elements

Use the left toolbar to add items:

- **Add Map** — draw a rectangle; your map appears inside it
- **Add Title** — click *Add Label*, type your title, make it large and bold
- **Add Legend** — shows what each



Exporting Your Map as PDF or PNG

Export Options

In the Print Layout, go to the **Layout** menu:

- **Export as PDF**
  - Best for **printing** and **formal reports**
  - Vector quality — sharp at any zoom level
  - Recommended resolution: 300 DPI
- **Export as Image (PNG)**
  - Best for **presentations** and **web use**
  - Fixed pixel size
  - Recommended resolution: 150–300 DPI

Step-by-Step Export

1. Ensure your layout looks correct
2. Go to **Layout** → **Export as PDF...**
3. Choose a save location and file name
4. Set resolution to **300 DPI**
5. Click **Save**
6. Open the PDF to verify — it should look crisp and professional

Save Your Project

Before exporting, save your QGIS project (**Ctrl + S**). This saves all layer connections

Session Summary

Map Making Workflow in QGIS

1. **Load data** — connect to CRD via WMS/WFS (Session 1)
2. **Style layers** — categorised or graduated symbols to show meaning
3. **Explore data** — use the attribute table to filter and select features
4. **Add labels** — show ward names, facility names on the map
5. **Create print layout** — add title, legend, scale bar, north arrow
6. **Export** — save as PDF for reports or PNG for presentations

Practical Exercise (Next)

You will now create your own professional map:

Key Shortcuts

- **Ctrl+S** — Save project
- **F5** — Refresh map